

ESMValTool

Earth System Model Evaluation Tool

An Introduction to ESMValTool

LST CCI Workshop, UK Met Office

May 14th, 2026

Ranjini Swaminathan¹, Birgit Hassler², Axel Lauer², Rob King³, Emma Hogan³, Bouwe Andela⁴ and Claire Bulgin¹

¹University of Reading and National Centre for Earth Observation, ² DLR,

³Met Office and ⁴Netherlands eScience Centre

Motivation

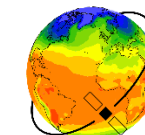
→ **Earth System Models (ESMs)** are important tools for

- improving our understanding of the climate system
- projecting future climate change

→ The **evaluation** of earth system models with earth observations (satellite, airborne...) is crucial for

- model improvements,
- a better process understanding of the climate system and
- more trustworthy climate projections are to be used for policy guidance.

**Challenge: Climate models
increase in complexity and resolution**



ESMs are Complex – and there are Plenty!

Latest Assessment Report of the IPCC (AR6):

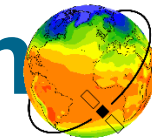
- ~60 ESMs analyzed
- >30 international institutions participated

→ **Coordination** necessary!

Institution Full Country or Region Name	Models	Main References	Atmosphere 1) Component Name 2) Resolution (km) and Number of Levels (L) 3) Top	Aerosol 1) Component Name 2) Emissions- driven or Prescribed 3) References	Atmospheric Chemistry 1) Component Name 2) Details 3) References	Ocean 1) Component Name 2) Horizontal Resolution and Number of Levels 3) Vertical Grid	Cryosphere 1) Sea Ice 2) Land Ice	Land 1) Component Name 2) Reference	Land Carbon Active Processes	Ocean Interactive Biogeochemistry 1) Component Name 2) Reference		
CAS Chinese Academy of Sciences China	Institution Full Country or Region Name	Models	Main References	Atmosphere 1) Component Name 2) Resolution (km) and Number of Levels (L) 3) Top	Aerosol 1) Component Name 2) Emissions- driven or Prescribed 3) References	Atmospheric Chemistry 1) Component Name 2) Details 3) References	Ocean 1) Component Name 2) Horizontal Resolution and Number of Levels 3) Vertical Grid	Cryosphere 1) Sea Ice 2) Land Ice	Land 1) Component Name 2) Reference	Land Carbon Active Processes	Ocean Interactive Biogeochemistry 1) Component Name 2) Reference	
CAS	CHRM and CERFACS	Institution Full Country or Region Name	Models	Main References	Atmosphere 1) Component Name 2) Resolution (km) and Number of Levels (L) 3) Top 4) References	Aerosol 1) Component Name 2) Emissions- driven or Prescribed 3) References	Atmospheric Chemistry 1) Component Name 2) Details 3) References	Ocean 1) Component Name 2) Horizontal Resolution and Number of Levels 3) Vertical Grid 4) References	Cryosphere 1) Sea Ice 2) Land Ice	Land 1) Component Name 2) Reference	Land Carbon Active Processes	Ocean Interact Biogeochemis t 1) Component Name 2) Reference
CCMA Canadian Centre for Climate Modelling and Analysis Canada	CSIRO Commonwealth Scientific and Industrial Research Organisation Australia	FIO-ONLM First Institute of Oceanography and Pilot National Laboratory for Marine Science and Technology (Qingdao), China	FIO-ESM-2-0	Bao et al. (2020)	1) CAM5 2) 100km, 26 L 3) Top 43 km	2) Prescribed, MACv2-SP (Stevens et al., 2017)	None	POP-W with MASNUM surface wave model 2) 60 km, 60 L 3) z 4) Qiao et al. (2013)	1) CICE4.0 2) Hunke and Lipscomb (2010)	1) CLIM4.0 2) Lawrence et al. (2011)	Land carbon N cycle	BEC
CCCR-ITM Centre for Climate Change Research, Indian Institute of Tropical Meteorology India	CCSR-ARCCSS CSIRO and Austr. Res. Council Centre of Excellence for Climate System Science Australia	FIO-ONLM First Institute of Oceanography and Pilot National Laboratory for Marine Science and Technology (Qingdao), China	FIO-ESM-2-0	Bao et al. (2020)	1) CAM5 2) 100km, 26 L 3) Top 43 km	2) Prescribed, MACv2-SP (Stevens et al., 2017)	None	POP-W with MASNUM surface wave model 2) 60 km, 60 L 3) z 4) Qiao et al. (2013)	1) CICE4.0 2) Hunke and Lipscomb (2010)	1) CLIM4.0 2) Lawrence et al. (2011)	Land carbon N cycle	BEC
CMCC Centro Euro- Mediterraneo sui Cambiamenti Climatici Italy	E3SM National Laboratories Consortium USA	HAMMOZ- Consortium Switzerland, Germany, UK, Finland	MPI-ESM-1-2-HAM	Neubauer et al. (2019a)	1) ECHAM6.3 2) 170 km, 47 L 3) Top 80 km	1) HAM2.3 2) Emissions-driven 3) Tegen et al. (2019)	2) Specified oxidants, sulphur chemistry 3) Feichter et al. (1996); Inness et al. (2013)	1) MPIOM 1.63 2) 100 km, 40 L 3) z	2) Notz et al. (2013)	1) JSBACH3.20 2) Reick et al. (2021)	Land carbon N cycle Prog. veg. Fires	HAMOC6
CNRM Centre National de Recherches Météorologiques and CERFACS Centre Européen de Recherche et de Formation Avancée en Calcul Scientifique France	EC-Earth Consortium Europe	INM Institute for Numerical Mathematics Russian Federation	INM-CM4-8 INM-CM5-0	INM-CM4-8: (Volodin et al., 2018) INM-CM5-0: (Volodin et al., 2017)	CM4: 1) INM-AM4-8 2) 150 km, 21 L 3) Top 31 km CM5: 1) INM-AM5.0 2) 150 km, 73 L 3) Top 61 km	1) INM-AER1 2) Emissions-driven 3) Volodin and Kostyrkin (2016)	None	INM-CM5 2) CM4: 70 km, 40 L CM5: 30 km, 40 L 3) Sigma 4) Zalesny et al. (2010)	1) INM-ICE1 2) Yakovlev (2009)	INM-LND1	Land carbon	None
	IPSL Institut Pierre- Simon Laplace France	IPSL Institut Pierre- Simon Laplace France	IPSL-CM6A-LR	Boucher et al. (2020g)	1) LMDZ NPv6 2) 160 km, 79 L 3) Top 80 km 4) (Hourdin et al., 2020)	2) Prescribed 3) Lurton et al. (2020)	2) Specified oxidants for aerosols	1) NEMO 3.6 2) 70 km, 75 L 3) z	1) NEMO-LIM3 2) Rousset et al. (2015)	ORCHIDEE (v2.0, Water/carbon/ energy mode)	None	PISCES
	IPSL	IPSL Institut Pierre- Simon Laplace France	IPSL-CM5A2-INCA		1) LMDZ APv5 2) 240 km, 79 L 3) Top 80 km 4) (Hourdin et al., 2020)	1) INCA 2) Emissions-driven	1) INCA 2) Interactive 3) Hauglustaine et al. (2014)	1) NEMO 3.6 2) 150 km, 30 L 3) z	1) NEMO-LIM3 2) Rousset et al. (2015)	ORCHIDEE (IPSL-CM5A2.1, Water/ carbon/energy mode)	Land carbon	PISCES
	KIOST Korea Institute of Ocean Science & Technology Republic of Korea	KIOST Korea Institute of Ocean Science & Technology Republic of Korea	KIOST-ESM	Pak et al. (2021)	1) GFDL-AM2.0 2) 190 km, 32 L 3) Top 43 km 4) Anderson et al. (2004)	1) GFDL-AM2.0 2) Emissions-driven 3) Anderson et al. (2004)	None	1) GFDL-MOMS.0 2) 90 km, 52 L 3) z	1) GFDL-SIS	GFDL-LM3.0 (Milly et al., 2014)	Land carbon N cycle Prog. veg.	TOPAZ2

IPCC AR6, Annex 2

CMIP – Coupled Model Intercomparison Project



ESMValTool
Earth System Model Evaluation Tool

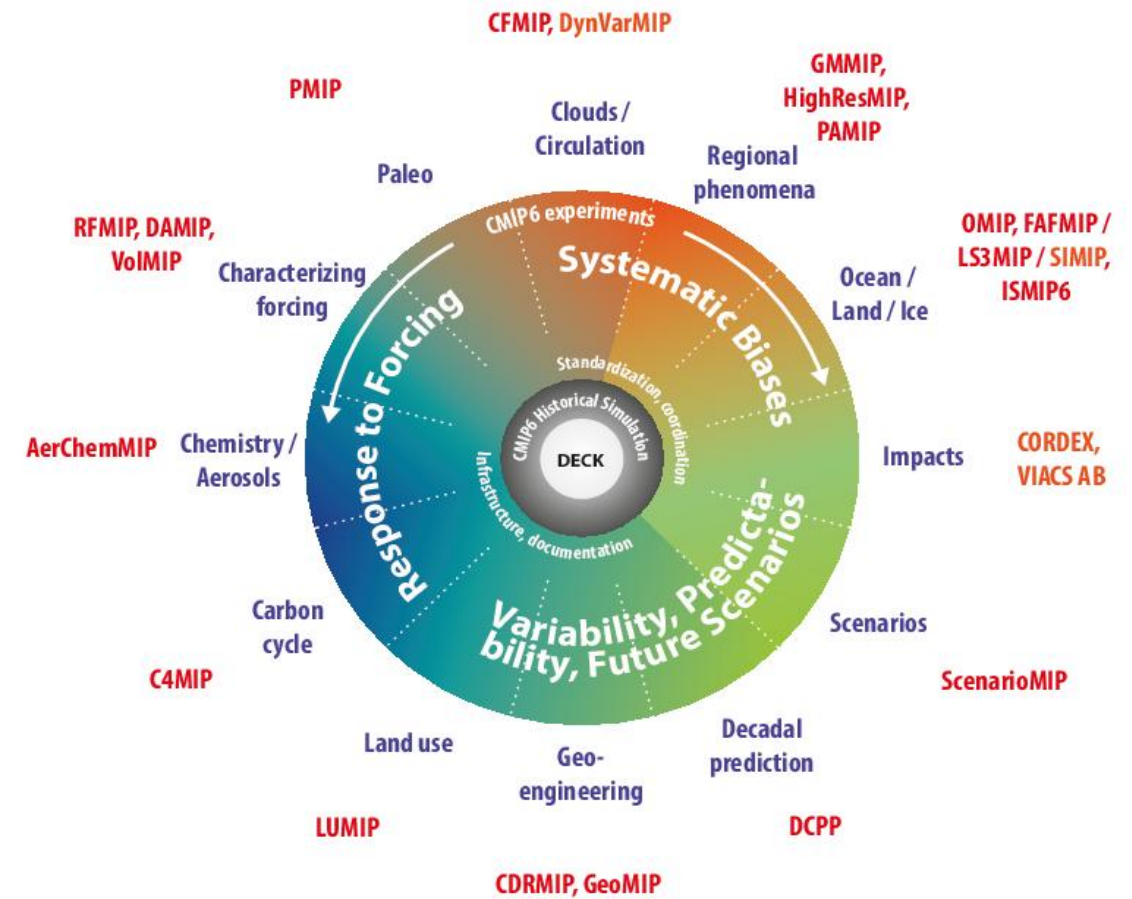
Aim of CMIP: Better understand past, present and future climate changes arising from natural and anthropogenic forcing in a multi-model context.

1) A handful of common experiments

- i. AMIP simulation (~1979-2014)
- ii. Pre-industrial control simulation
- iii. 1%/year CO₂ increase
- iv. Abrupt 4xCO₂ run
- v. Historical simulation using CMIP6 forcings (1850-2014)

2) Standardization, coordination, infrastructure, documentation

3) CMIP-Endorsed Model Intercomparison Projects (MIPs)



Eyring et al., GMD (2016)

ESMValTool

We have:

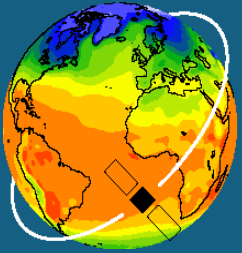
Many complex CMIP models with standardized output

Crucial question:

How well do these models perform relative to reference data sets (e.g., observations)? → **Standardized model evaluation**



ESMValTool in a nutshell

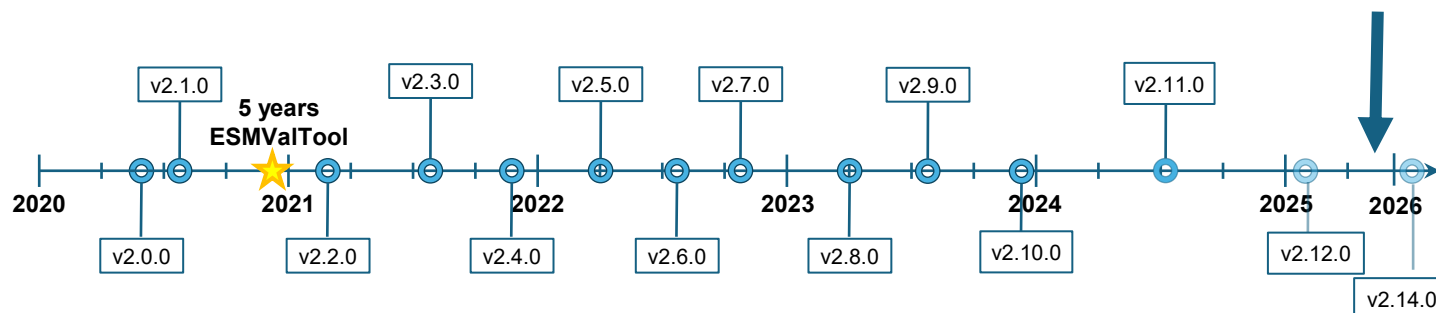


ESMValTool

Earth System Model Evaluation Tool

- Community diagnostic and performance metrics tool for **evaluation and analysis of Earth system models**, primarily focused on multi-model context
- Open source community development on GitHub** (> 200 developers, > 60 international institutes, funded by many national and international projects)
- Used in several chapters of the **Assessment Report 6** of the IPCC's WG1 and provenance records available for all results
- Release of v2.0.0 in August 2020, currently at v2.14.0

V2.13.0 and 10th anniversary!!!



Scientific Documentation

Righi et al., GMD, 2020

Technical overview

Eyring et al., GMD, 2020

Large-scale diagnostics

Lauer et al., GMD, 2020

Diagnostics for emergent constraints and future projections

Weigel et al., GMD, 2021

Diagnostics for extreme events, regional and impact evaluation

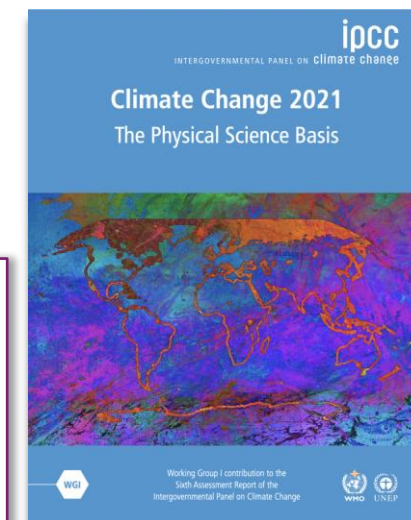
Schlund et al., GMD, 2023

Evaluation of native ESM output

Lauer et al., GMD, 2025

Monitoring and benchmarking

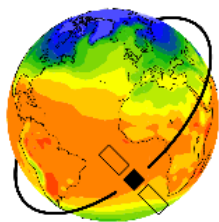
- ~ 60 scientific peer-reviewed publications
- > 20 national and international science projects, including large EU projects
- ESMValTool has been used for the production of figures for **several chapters of the Assessment Report 6 of the IPCC's WG1** (including provenance record)



*Eyring et al., Chapter 3 of
IPCC AR6 WGI, 2021*

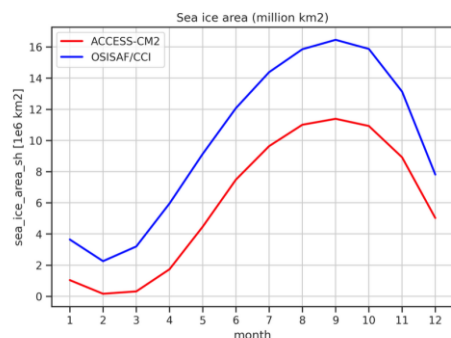
ESMValTool and Climate-REF

- Rapid Evaluation Framework (REF).
- ESMValTool - one of four open-source software providers.
- Diagnostics include – climate drivers of fire, climate at global warming levels, cloud properties, ENSO and climate sensitivity.
- Explore : <https://dashboard.climate-ref.org/>



ESMValTool

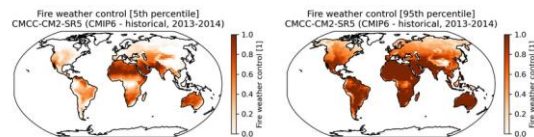
Earth System Model Evaluation Tool



20-year average seasonal cycle of the sea ice area in million km² from ACCESS-CM2.r10i1p1f1.gn compared with OSI-450.

Execution Group: cmip6_gn_r10i1p1f1_ACCESS-CM2

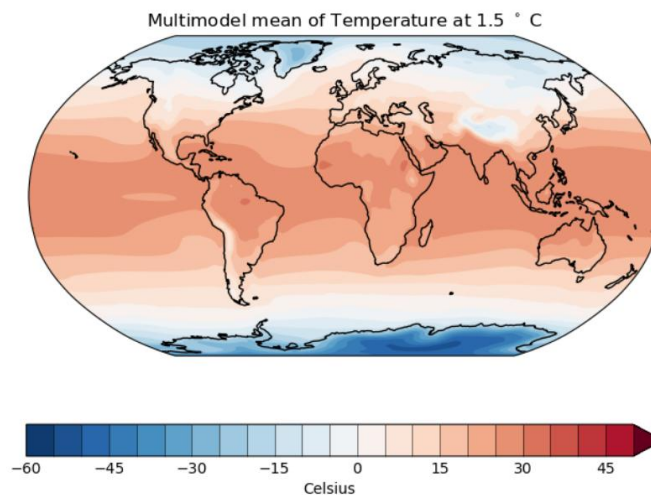
Filename: executions/recipe_20251003_125750/plots/siarea_



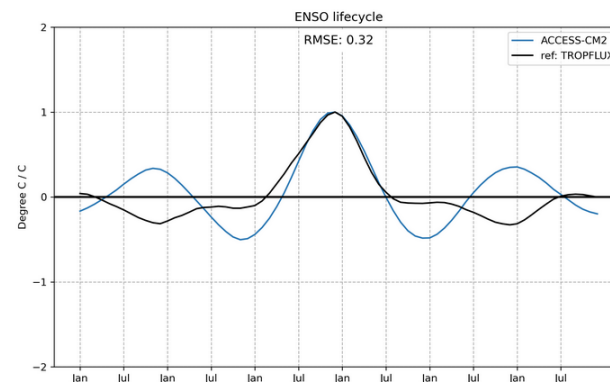
Fire weather control for the CMCC-CM2-SR5 (CMIP6-historical) for the time period 2013/2014 as computed with the ConFire model (Jones et al., 2024).

Execution Group: cmip6_gn_r10i1p2f1_CMCC-CM2-SR5

Filename: executions/recipe_20251002_035023/plots/fire_ev



Multimodel mean of Temperature at 1.5 °C



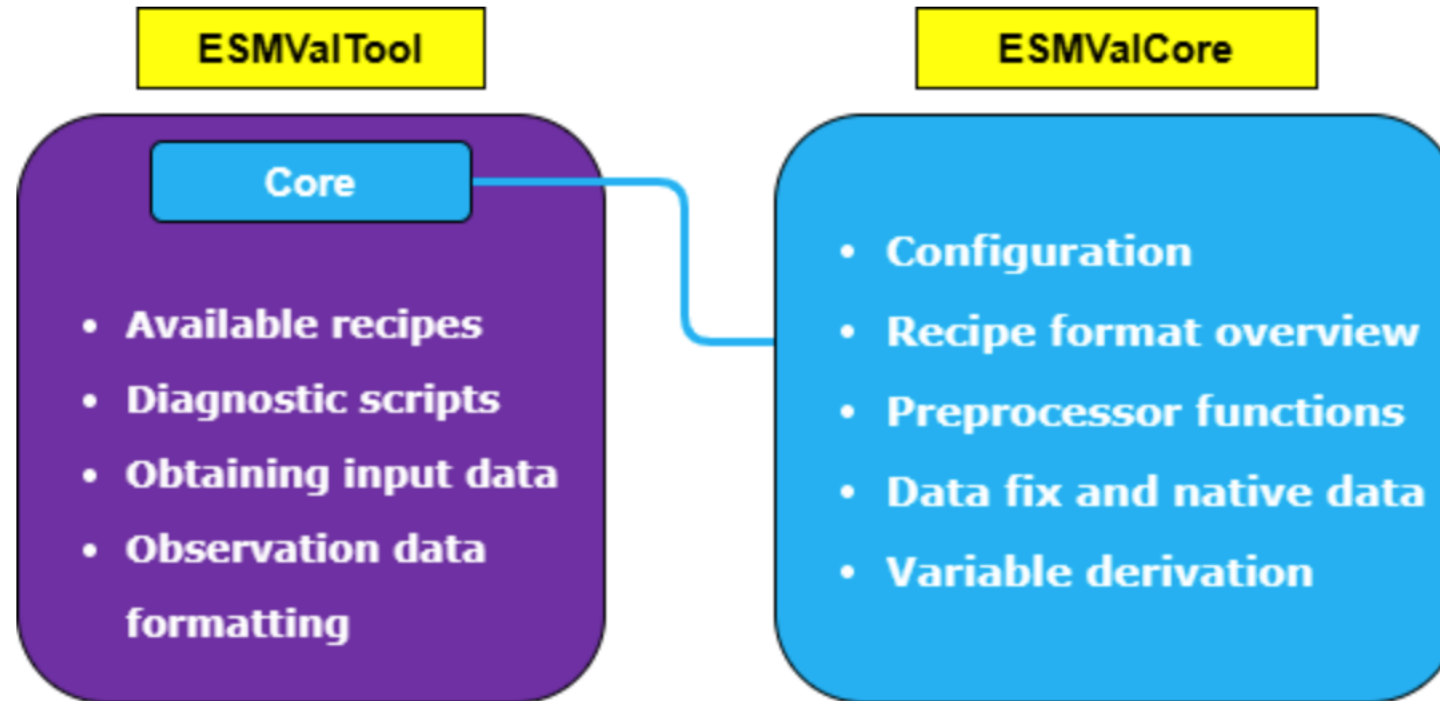
Temporal evolution of sea surface temperature anomalies in the central equatorial Pacific (Niño 3.4 region average), illustrating the ENSO-associated variability.

Execution Group: cmip6_gn_r10i1p1f1_ACCESS-CM2

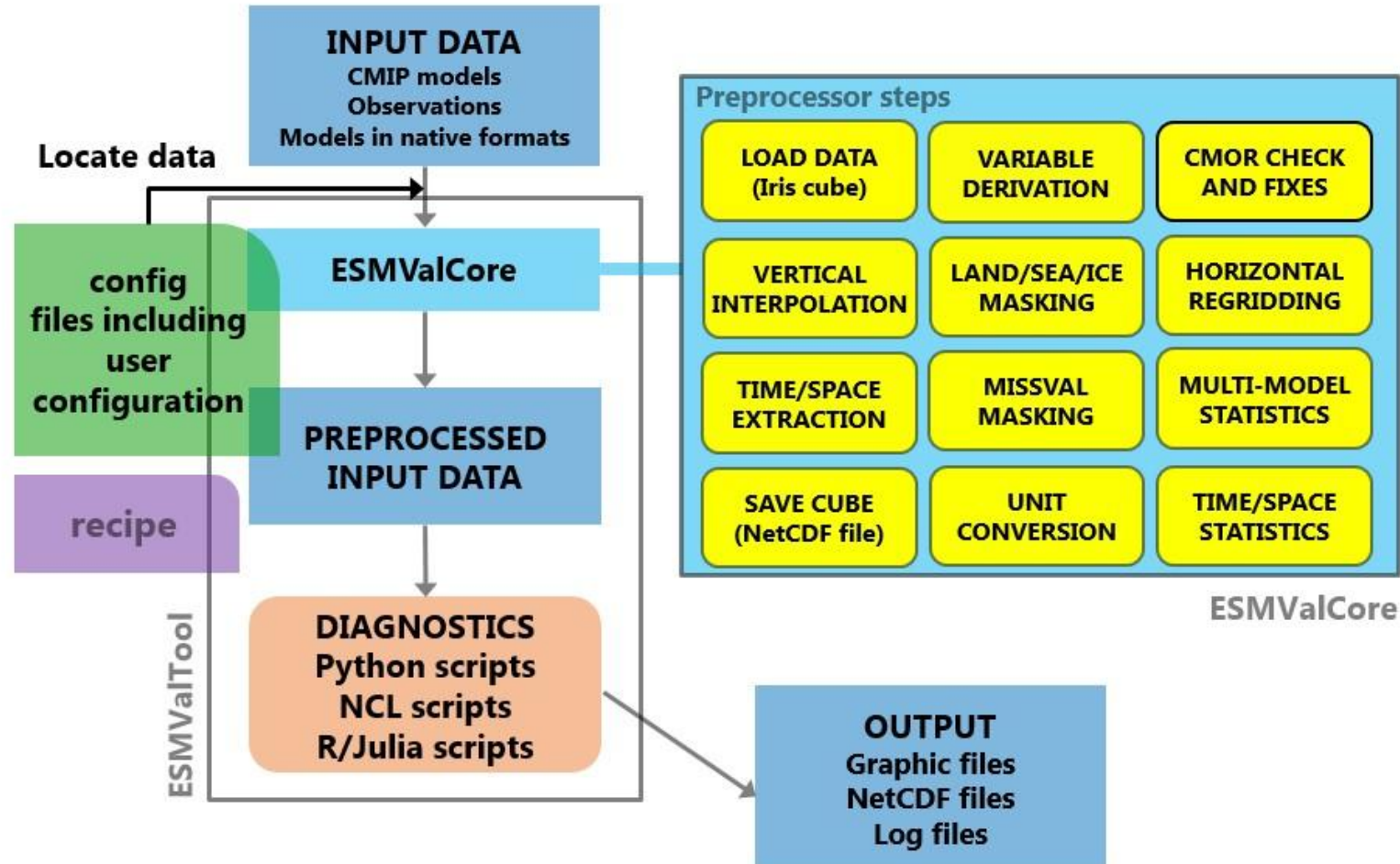
Filename: executions/recipe_20251002_035527/plots/diagno

ESMValTool and Climate-REF

ESMValTool – Components



ESMValTool – Details

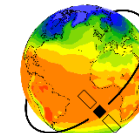


ESMValTool Recipes

ESMValTool is controlled through a **recipe** (text file in YAML format).

Recipes contain

- Datasets to analyze
- Preprocessors to use
- Variables to analyze
- Diagnostic scripts to use (Python, NCL or R scripts)



ESMValTool Recipes

datasets:

- {dataset: ERA5, project: OBS6, tier: 3, type: reanaly, version: 1}
- {dataset: CanESM5, project: CMIP6, exp: historical, grid: gn, ensemble: r1i1p1f1}

preprocessors:

regridding_step:

regrid:

target_grid: 1x1

scheme: linear

diagnostics:

example:

variables:

tas:

mip: Amon

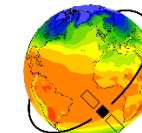
timerange: '1980/2005'

preprocessor: regridding_step

scripts:

example_script:

script: ~/diag_scripts/example.py



ESMValTool Recipes

datasets:

- {dataset: ERA5, project: OBS6, tier: 3, type: reanaly, version: 1}
- {dataset: CanESM5, project: CMIP6, exp: historical, grid: gn, ensemble: r1i1p1f1}

preprocessors:

```
regridding_step:  
  regrid:  
    target_grid: 1x1  
    scheme: linear
```

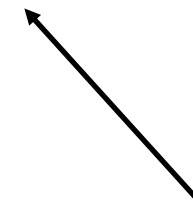
diagnostics:

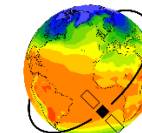
```
example:  
  variables:  
    tas:  
      mip: Amon  
      timerange: '1980/2005'  
      preprocessor: regridding_step
```

scripts:

```
example_script:  
  script: ~/diag_scripts/example.py
```

All datasets





ESMValTool Recipes

datasets:

- {dataset: ERA5, project: OBS6, tier: 3, type: reanaly, version: 1}
- {dataset: CanESM5, project: CMIP6, exp: historical, grid: gn, ensemble: r1i1p1f1}

preprocessors:

regridding_step:

regrid:

target_grid: 1x1

scheme: linear

Preprocessor
settings

diagnostics:

example:

variables:

tas:

mip: Amon

timerange: '1980/2005'

preprocessor: regridding_step

scripts:

example_script:

script: ~/diag_scripts/example.py

ESMValTool Recipes

datasets:

- {dataset: ERA5, project: OBS6, tier: 3, type: reanaly, version: 1}
- {dataset: CanESM5, project: CMIP6, exp: historical, grid: gn, ensemble: r1i1p1f1}

preprocessors:

regridding_step:

regrid:

target_grid: 1x1

scheme: linear

diagnostics:

example:

variables:

tas:

mip: Amon

timerange: '1980/2005'

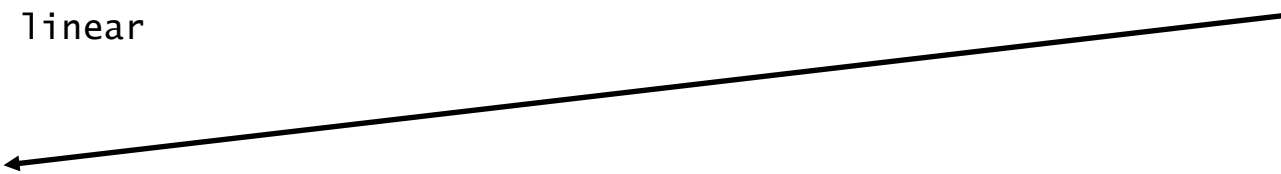
preprocessor: regridding_step

scripts:

example_script:

script: ~/diag_scripts/example.py

Diagnostic scripts



ESMValTool Recipes

datasets:

- {dataset: ERA5, project: OBS6, tier: 3, type: reanaly, version: 1}
- {dataset: CanESM5, project: CMIP6, exp: historical, grid: gn, ensemble: r1i1p1f1}

preprocessors:

regridding_step:

regrid:

target_grid: 1x1
scheme: linear

diagnostics:

example:

variables:

tas:

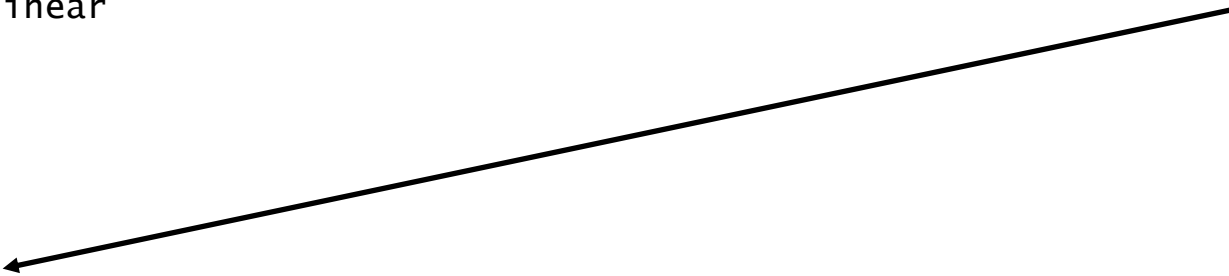
mip: Amon
timerange: '1980/2005'
preprocessor: regridding_step

scripts:

example_script:

script: ~/diag_scripts/example.py

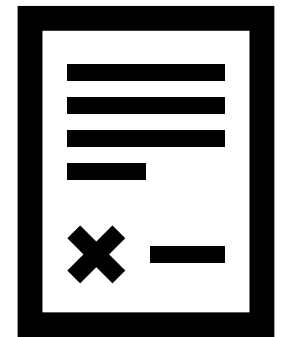
Variables



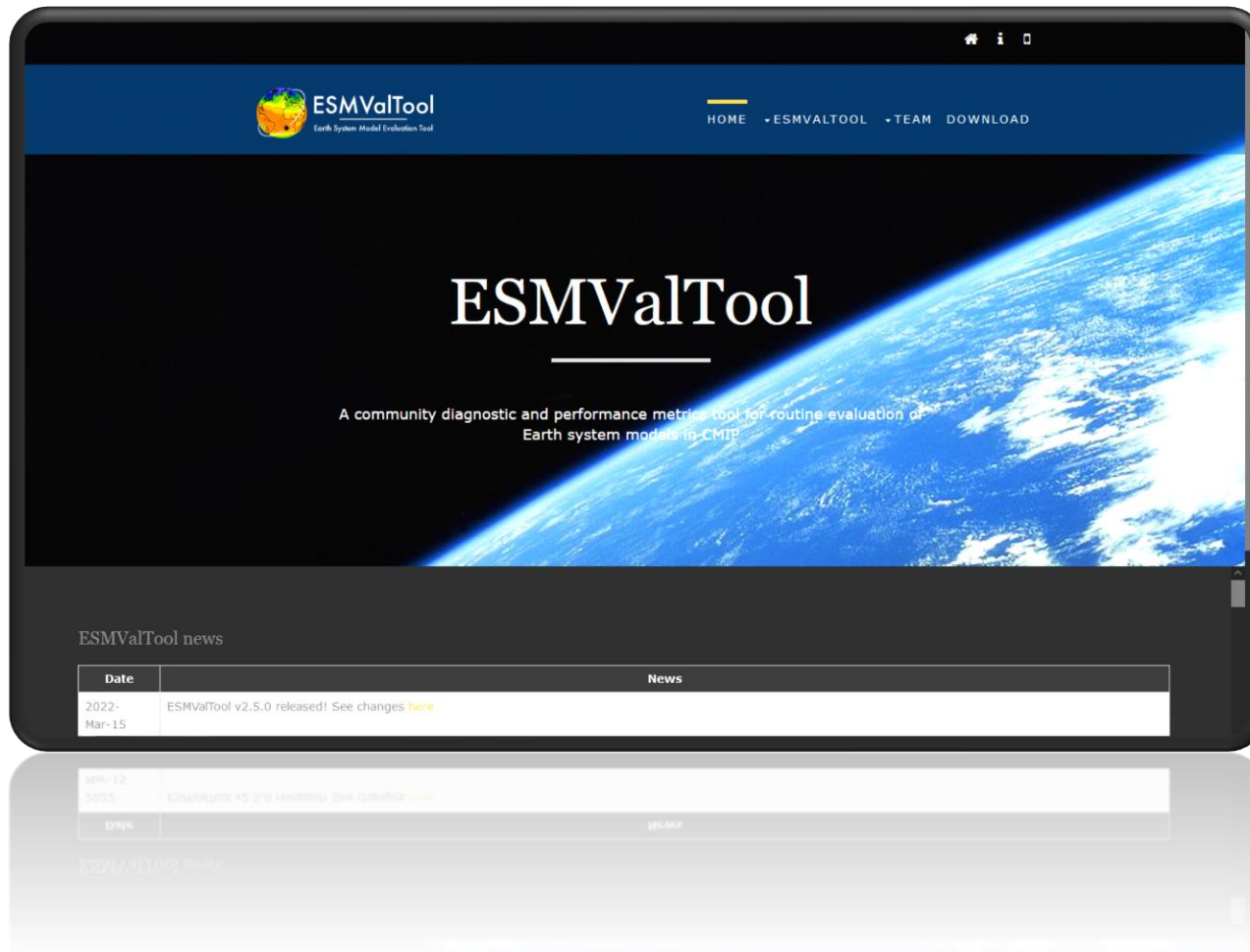
ESMValTool Consortium

Agreement signed in **December 2024** by:

- DLR, Germany (Co-PI)
- Met Office, UK (Co-PI)
- Barcelona Supercomputing Center (BSC), Spain
- SMHI, Sweden
- NCAS and NCEO, UK
- NLeSC, Netherlands
- ACCESS-NRI – more recently



Website



General presentation of ESMValTool:

- Important links,
- gallery,
- license,
- presentations,
- meeting notes,...

<https://esmvaltool.org>

GitHub Issues

<https://github.com/ESMValGroup/ESMValTool/issues>

 [ESMValGroup](#) / [ESMValTool](#) Public

 Notifications  Fork 135  Star 244

[Code](#) [Issues](#) 293 [Pull requests](#) 98 [Discussions](#) [Actions](#) [Projects](#) 1 [Security](#) [Insights](#)

 **Replace `prospector` with `ruff` integrated in Codacy**
#3739 · valeriupredoi opened on Aug 16, 2024

 **'stable' documentation build is broken and still points to v2.7**
#3490 · bouweandela opened on Dec 20, 2023  3

 [Labels](#) [Milestones](#) [New issue](#)

Open 293 **Closed** 1.264

Author ▾ Labels ▾ Projects ▾ Milestones ▾ Assignees ▾ Types ▾  Newest ▾

 **RTW configuration issues** [Recipe Test Workflow \(RTW\)](#)
#4076 · ehogan opened 2 weeks ago ·  v2.13.0 

 **Collection of outdated Documentation parts**
#4074 · bettina-gier opened 2 weeks ago

 **RTW config variable `BRANCH` used in both Tool and Core**
#4069 · alistairsellar opened 2 weeks ago  1  1 

 **Use `ESMVALTOOL_CONFIG_DIR` rather than `--config_dir` in RTW** [Recipe Test Workflow \(RTW\)](#)
#4053 · ehogan opened 3 weeks ago 

Discussions

<https://github.com/ESMValGroup/ESMValTool/discussions>

ESMValGroup / ESMValTool Public

Notifications

Fork 135

Star 244

<> Code

Issues 293

Pull requests 98

Discussions

Actions

Projects 1

Security

Insights



Automated formatting

 Ideas · bouweandela

Q is:open

×

Sort by: Latest activity ▾

Label ▾

Filter: Open ▾

Categories

 View all discussions

 General

 Ideas

 New to ESMValTool

 Polls

Discussions

↑ 1



Query about using the custom CMOR variables options for ICON-ART

nchawang asked last week in [Q&A](#) · **Answered**

   4

↑ 1



Can we load ICON dust AOD with ESMValTool?

nchawang asked on Mar 31 in [Q&A](#) · **Answered**

   5



Exclude legacy recipes from gallery

Community Discussions

<https://github.com/ESMValGroup/Community/discussions>

ESMValGroup / Community Public

Notifications Fork 0 Star 5

<> Code Issues Pull requests 1 Discussions Actions Projects Security Insights

**ESMValTool long-term strategy**
General · hb326

**What technical advancements do we need for CMIP7?**
General · hb326

**Agenda ESMValTool Workshop May 13-15 Oberpfaffenhofen, Germany.**
Meetings · bettina-gier

is:open

Sort by: Latest activity

Label

Filter: Open

Categories

- View all discussions
- Announcements
- General
- Meetings
- Polls

Discussions

1

**Monthly Meeting: June**
monthly community meeting
bettina-gier started 2 weeks ago in Meetings

6

**Agenda ESMValTool Workshop May 13-15 Oberpfaffenhofen, Germany.**
workshop
bettina-gier started on Mar 7 in Meetings

Documentation

<https://docs.esmvaltool.org/>



[Introduction](#) [ESMValTool Functionalities](#) [Getting started](#) [Gallery](#) [Recipes](#) [More ▾](#)



Welcome to ESMValTool's documentation!

To get a first impression of what ESMValTool and ESMValCore can do for you, have a look at our blog posts [Analysis-ready climate data with ESMValCore](#) and [ESMValTool: Recipes for solid climate science](#).

A tutorial is available on <https://tutorial.esmvaltool.org>.

A series of video lectures has been created by [ACCESS-NRI](#). While these are tailored for ACCESS users, they are still very informative.



On this page

Welcome to ESMValTool's documentation!

[Indices and tables](#)

This Page

- [Show Source](#)

  **latest** ▾

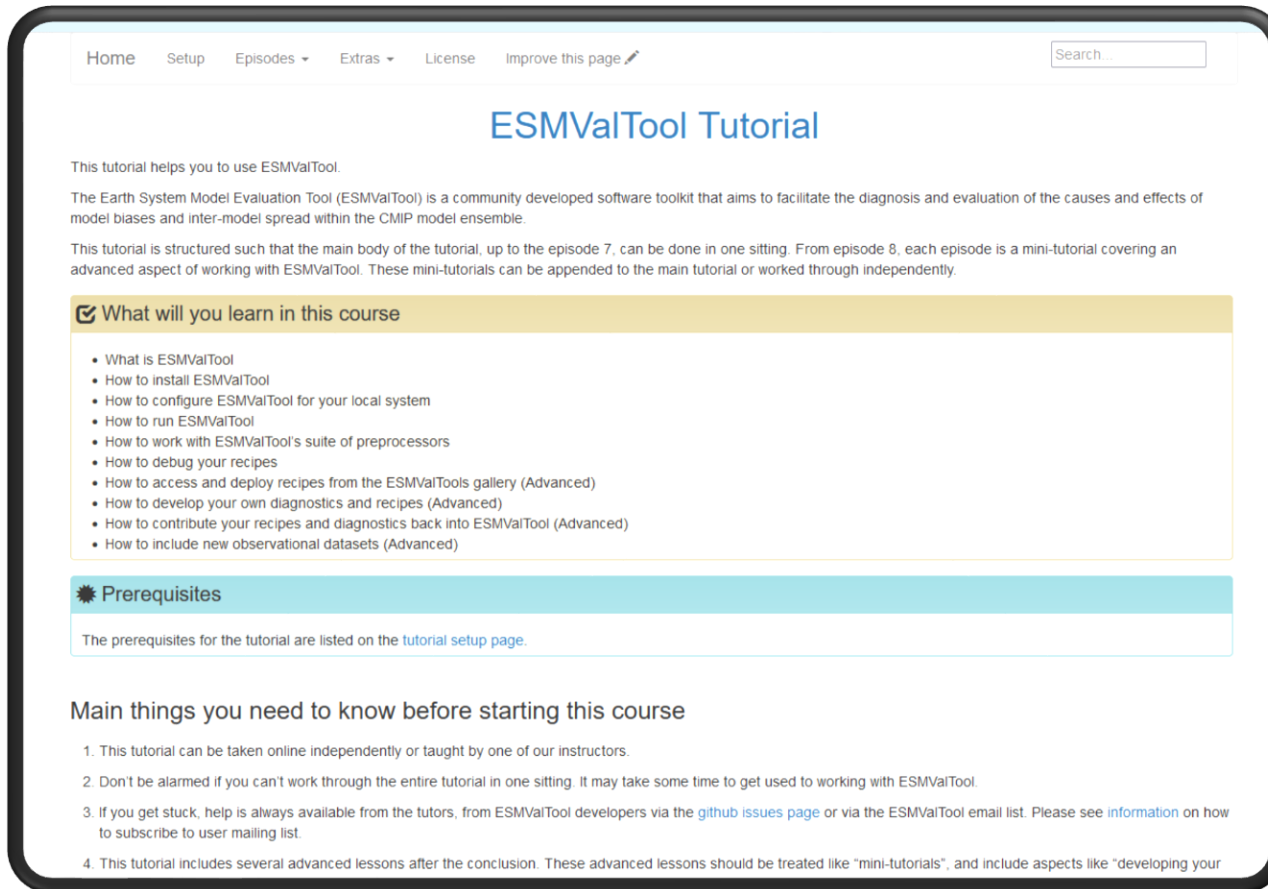
Tutorial

<https://tutorial.esmvaltool.org>

→ basic and an advanced tutorial

User Engagement Team Tasks:

- Maintaining tutorial on GitHub
- Keeping the tutorial updated with the releases
- Handling with user feedback
- Contact for tutorial hosts



The screenshot shows the ESMValTool Tutorial website. At the top is a navigation bar with links: Home, Setup, Episodes, Extras, License, and Improve this page. A search bar is on the right. The main heading is "ESMValTool Tutorial". Below it, a paragraph states: "This tutorial helps you to use ESMValTool. The Earth System Model Evaluation Tool (ESMValTool) is a community developed software toolkit that aims to facilitate the diagnosis and evaluation of the causes and effects of model biases and inter-model spread within the CMIP model ensemble. This tutorial is structured such that the main body of the tutorial, up to the episode 7, can be done in one sitting. From episode 8, each episode is a mini-tutorial covering an advanced aspect of working with ESMValTool. These mini-tutorials can be appended to the main tutorial or worked through independently."

What will you learn in this course

- What is ESMValTool
- How to install ESMValTool
- How to configure ESMValTool for your local system
- How to run ESMValTool
- How to work with ESMValTool's suite of preprocessors
- How to debug your recipes
- How to access and deploy recipes from the ESMValTools gallery (Advanced)
- How to develop your own diagnostics and recipes (Advanced)
- How to contribute your recipes and diagnostics back into ESMValTool (Advanced)
- How to include new observational datasets (Advanced)

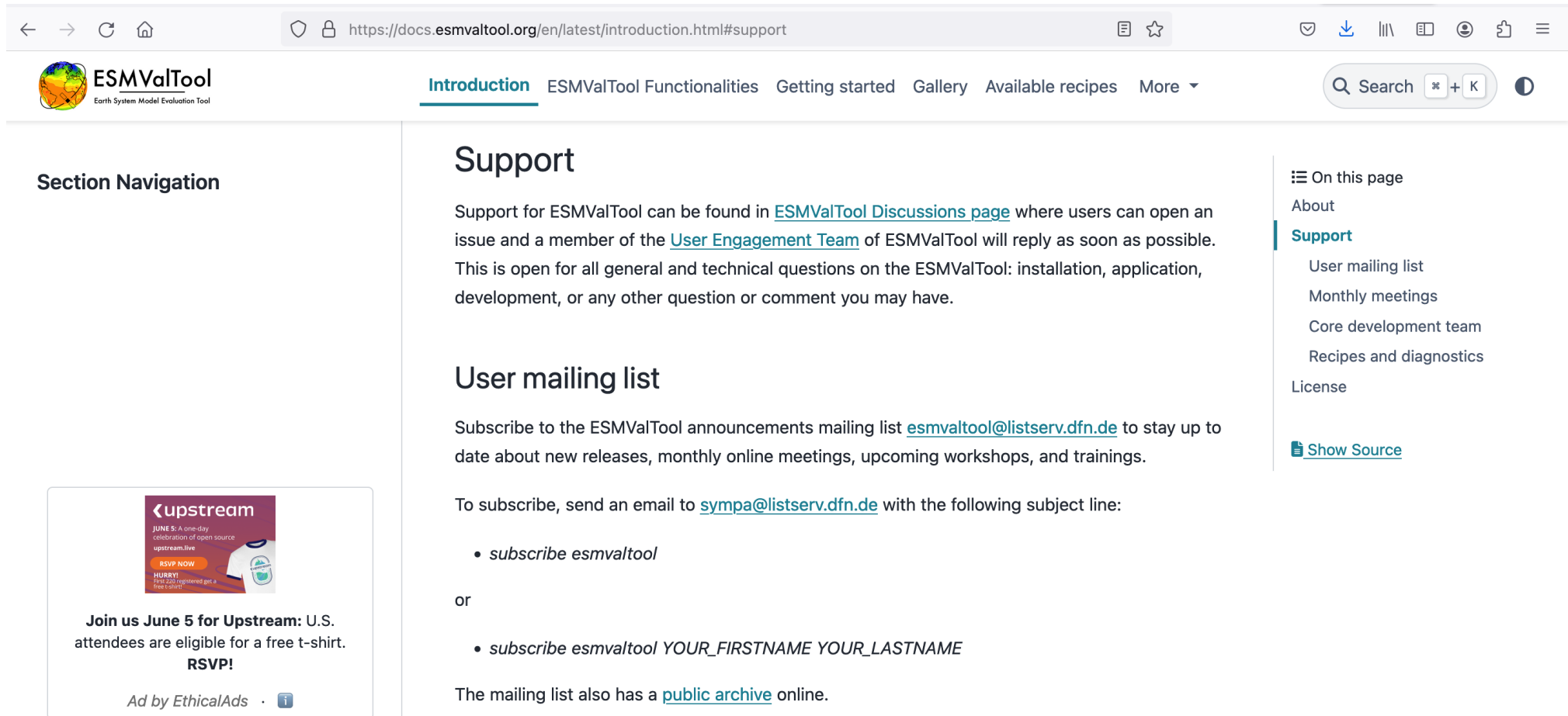
Prerequisites

The prerequisites for the tutorial are listed on the [tutorial setup page](#).

Main things you need to know before starting this course

1. This tutorial can be taken online independently or taught by one of our instructors.
2. Don't be alarmed if you can't work through the entire tutorial in one sitting. It may take some time to get used to working with ESMValTool.
3. If you get stuck, help is always available from the tutors, from ESMValTool developers via the [github issues page](#) or via the ESMValTool email list. Please see [information](#) on how to subscribe to user mailing list.
4. This tutorial includes several advanced lessons after the conclusion. These advanced lessons should be treated like "mini-tutorials", and include aspects like "developing your

Questions?



Thanks!

